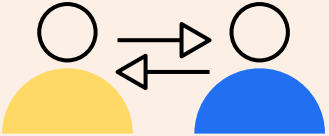
Transaction Risk & Escrow Recommendation

Problem & What We Want to Solve



Problem

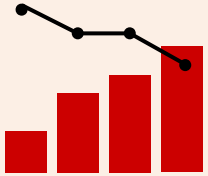
Customers abuse the return system

Department

The company has **no dedicated fraud detection department** to check these cases.

Manual Checks

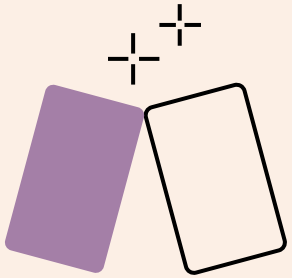
Manual checks cause **time loss** and are **error-prone**.



Financial Losses

This leads to **financial losses** and **loss of customer trust**.

Automated Fraud Risk Scoring System



Analyze

Automatically analyze transactions to **flag high-risk cases**.

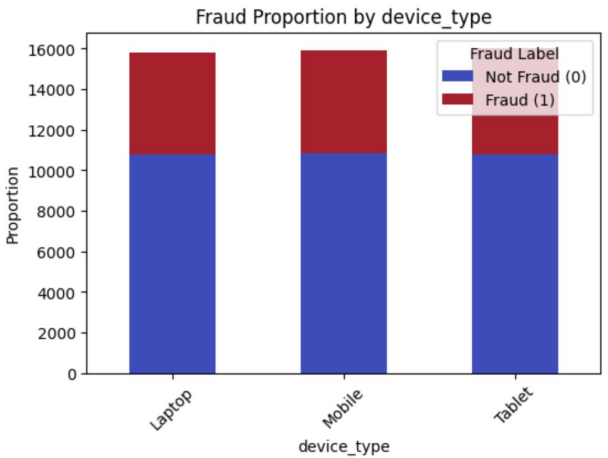
Recommend

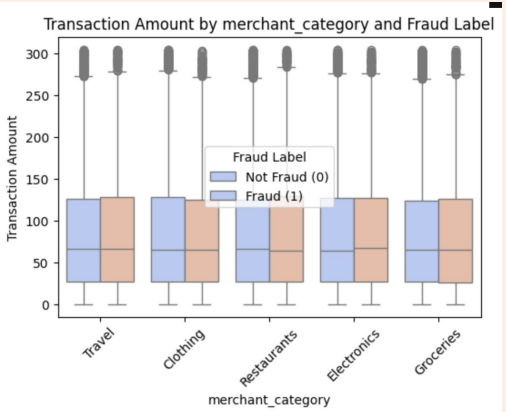
Recommend **Escrow** (secure payment service) for those risky cases

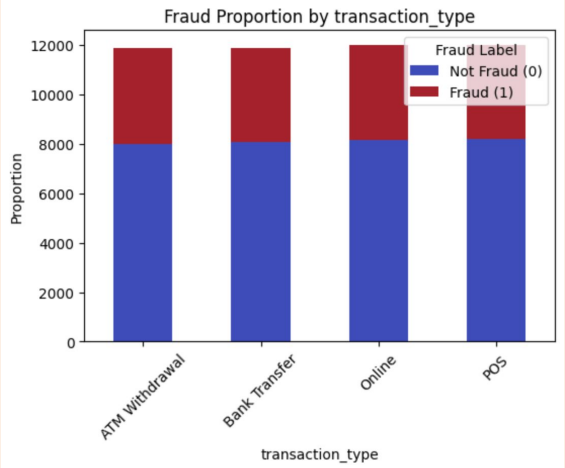
So That

- The company doesn't lose the sale,
- Fraud risk is minimized,
- Less manual effort is required — the system does the checks automatically.

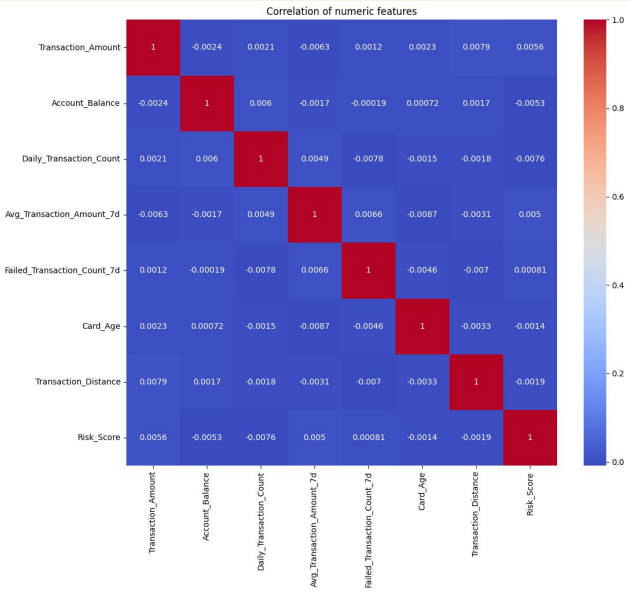
From Data to Features

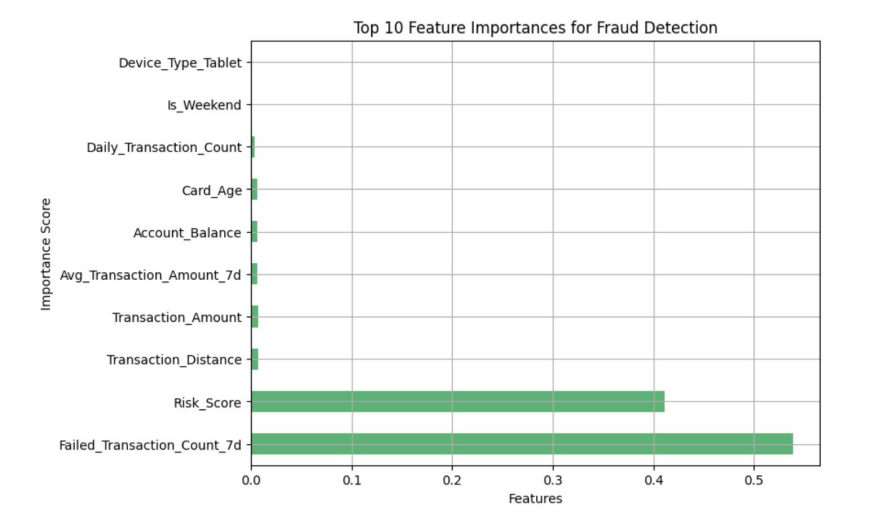






Where Does Risk Increase?





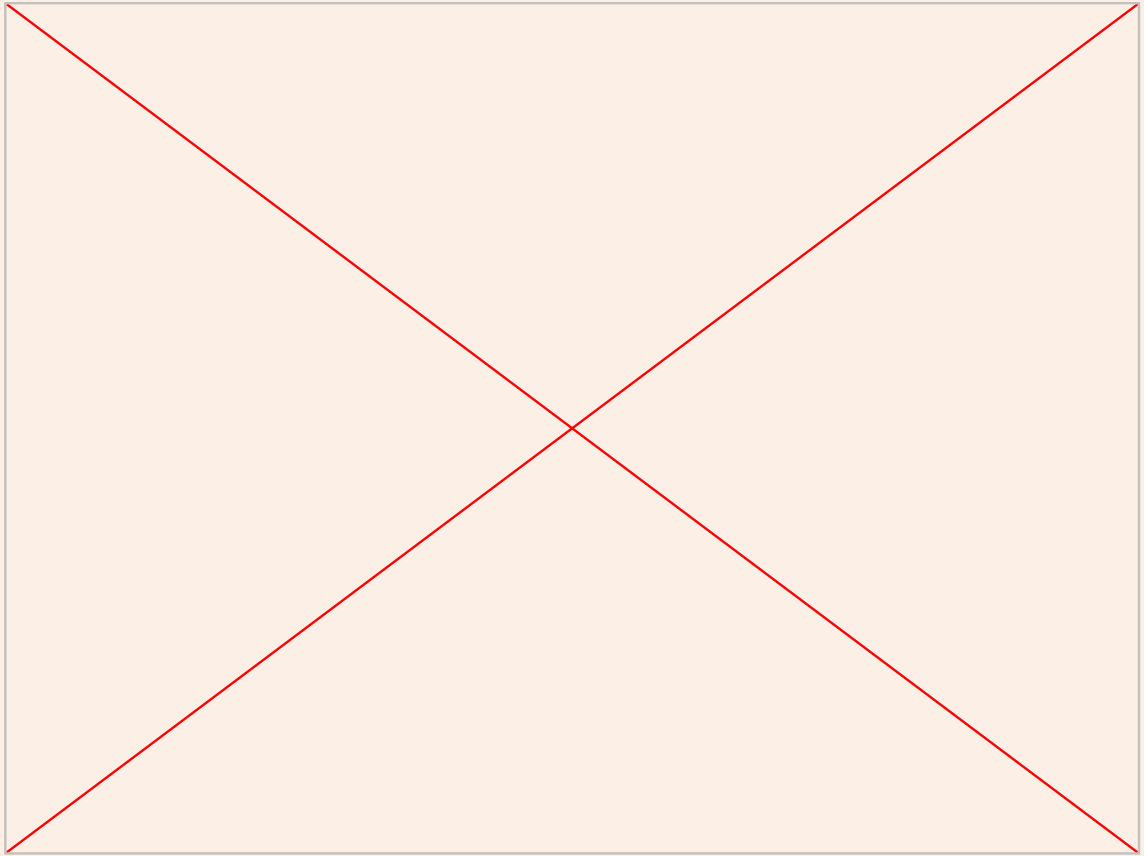
Model Testing & Evaluation

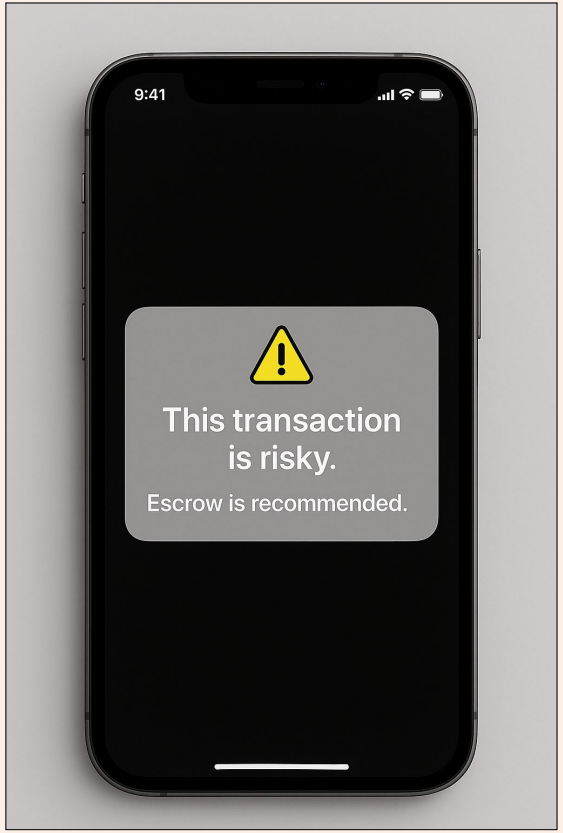
🔎 Transaction Risk & Escrow Recommendation

Escrow is recommended when the model flags **High Risk** or any domain rule is triggered.

	count	transaction_value	location_distance	platform_security	risk_score_model	model_high_risk	rule_flag	escrow_recommendation
17	518	high	near	low	0.87	1	0	⚠️ Escrow is Recommended
18	1,950	high	near	low	0.8	1	0	⚠️ Escrow is Recommended
19	1,351	low	near	low	0.9233	1	0	⚠️ Escrow is Recommended
20	214	low	near	high	0.0033	0	0	✅ Transaction seems Safe
21	964	high	near	low	0.2667	0	0	✅ Transaction seems Safe
22	60	low	near	low	0.18	0	0	✅ Transaction seems Safe
23	255	low	near	high	0	0	0	✅ Transaction seems Safe
24	1,128	low	far	low	0.0467	0	0	✅ Transaction seems Safe
25	1,329	low	far	low	0.83	1	0	⚠️ Escrow is Recommended
26	1,923	low	near	low	0.0767	0	0	✅ Transaction seems Safe
27	621	low	near	low	0.9233	1	0	⚠️ Escrow is Recommended

Download Scored Test (CSV)





Conclusion — Benefits of Our System

- Company keeps the sale
- Fraud risk is reduced
- Less manual work — system checks automatically

Thank you for your attention!